

Project Management Document

Project: BaSyx Editor Plugin Extension
Course: Software Engineering
Class: TINF24F
Professors: Markus Rentschler, Pawel Wojcik

Version History

Version	Date	Author	Comment
0.1	21.10.2025	Leonardo Risatti	Initial structure / Prototype
0.2	21.10.2025	Leonardo Risatti	Added content
0.3	23.10.2025	Leonardo Risatti	Small changes
1.0	19.11.2025	Leonardo Risatti	Major Rework of Pm

1. Project Assignment

Objective

The project extends the **Eclipse BaSyx UI** by developing an **Editor Plugin**, a **Viewer Plugin**, and enhancing the associated **REST API backend**.

It enables the structured import, validation, and integration of external engineering model files (e.g., KBL and VEC) into the **Asset Administration Shell (AAS)**.

The system provides automated plausibility checks, data extraction, and linking mechanisms, as well as a structured visualization of XML-based data within the BaSyx environment.

Additionally, this document establishes a common understanding of the system scope and serves as a foundation for development, testing, and validation throughout the project lifecycle.

Stakeholders

- Customer:** Markus Rentschler, Pawel Wojcik
- Users:** Developers and engineers working with AAS and engineering data
- Project Team:** Team 3

Main Tasks

- Requirements analysis and specification (Use Cases, functional & non-functional requirements)
- System design (architecture, interfaces, data flow)
- Implementation of editor and viewer plugins as well as backend extensions
- Testing and validation of all system components
- Documentation

Benefit

- Reduction of manual effort when integrating data
- Improved data quality and consistency through automated validation
- Enhanced usability and transparency within the BaSyx ecosystem
- Structured and user-friendly visualization of complex XML data
- Clear basis for implementation and verification via defined requirements

Deadlines

- First presentation: 21.11.2025
- Handover to stakeholders: 21.05.2026
- Final presentation: 22.05.2026

2. Project Organization

Role	Name
Project Director	Martin Böhm
Project Manager	Florian Zahn
Test Manager	Daniel Ziegler
System Architect	Federico Dibenedetto / Felix Bandl
Technical Documentation	Leonardo Risatti / Morten Haase

All team members contribute to development.

3. Project Context

This section describes the context in which the project is carried out. It outlines the organizational, technical, and temporal framework conditions as well as the involved stakeholders.

Temporal Context

Phase	Description
Pre-project	Familiarization with BaSyx architecture, AAS concepts, repository structure, and model formats (KBL/VEC)
Project phase	Iterative development of editor and viewer plugins, REST API extensions, and integration into the existing BaSyx system
Post-project	Handover to stakeholders, documentation finalization, and potential future extensions

Social Context

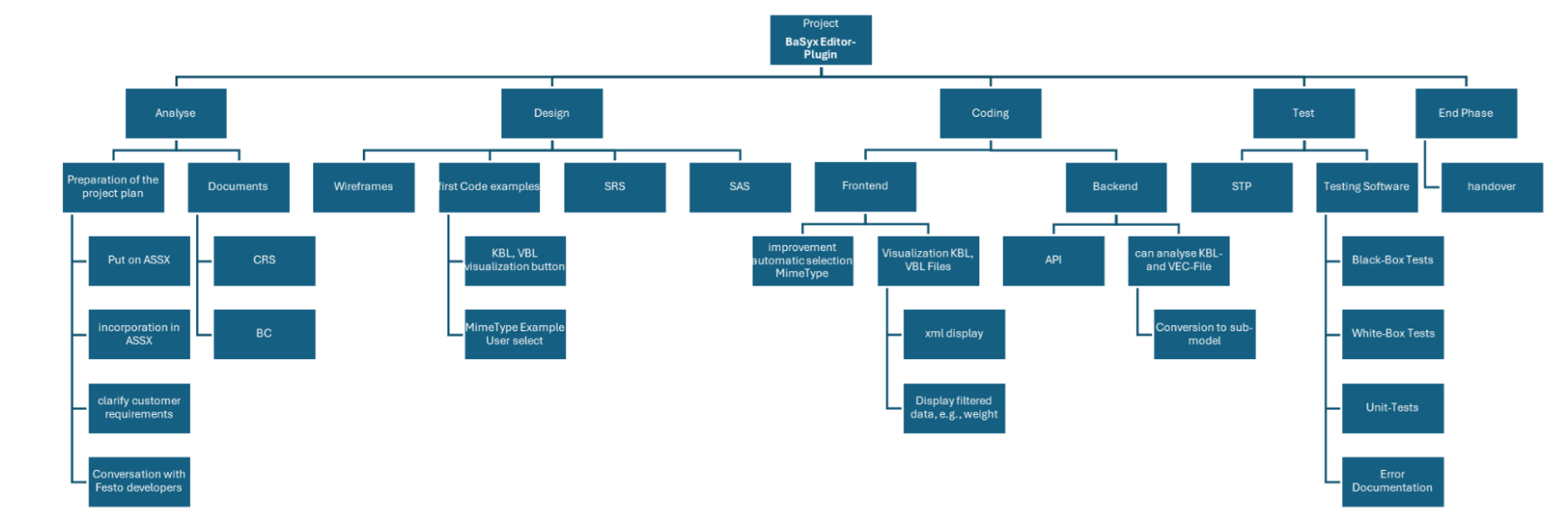
Stakeholder	Role	Opportunity	Risk	Mitigation
Customer	Define requirements, evaluate results	Improved teaching and demonstrability of BaSyx extensions	Changing or unclear requirements	Regular reviews and feedback sessions
Users	Use the system for data integration and analysis	More efficient workflows and better data understanding	Misinterpretation of features or workflows	User documentation, examples, and demos
Project Team	Design and implement the solution	Successful project delivery and technical learning	Overload, knowledge gaps	Clear task distribution, internal communication, documentation
Open-Source Community	Potential future users / contributors	Reusable extension for broader ecosystem	Compatibility issues with upstream changes	Modular design, alignment with BaSyx standards

Goals / Non-Goals

Goals	Non-Goals
Seamless integration of external model files (KBL, VEC) into AAS	Redesign of the complete BaSyx backend architecture
Automated validation, extraction, and linking of engineering data	Replacement of existing BaSyx core functionalities
Extension of REST API for structured XML data access	Support for arbitrary or unrelated data formats
Visualization of XML data in a structured and user-friendly way	Full UI redesign beyond plugin scope
Integration into existing plugin-based BaSyx architecture	Development of a standalone system outside BaSyx

4. Project Structure Plan(PSP)

The **Project Structure Plan (PSP)** organizes the project into five main phases: **Analysis, Design, Coding, Testing, and End Phase**, each consisting of clearly defined sub-tasks.



During the **Analysis phase**, the project plan is prepared and fundamental requirements are clarified. This includes the integration into ASSX, requirement discussions with stakeholders, and alignment with external partners. In parallel, essential project documents such as the CRS (Customer Requirements Specification) and Business Case (BC) are created.

The **Design phase** focuses on defining the system structure and user interaction. This includes the creation of wireframes, initial code examples, and technical specifications such as the Software Requirements Specification (SRS) and Software Architecture Specification (SAS).

In the **Coding phase**, the defined system functionalities are implemented based on the previously specified requirements and use cases. The focus lies on translating the designed concepts into a working software solution within the existing BaSyx architecture.

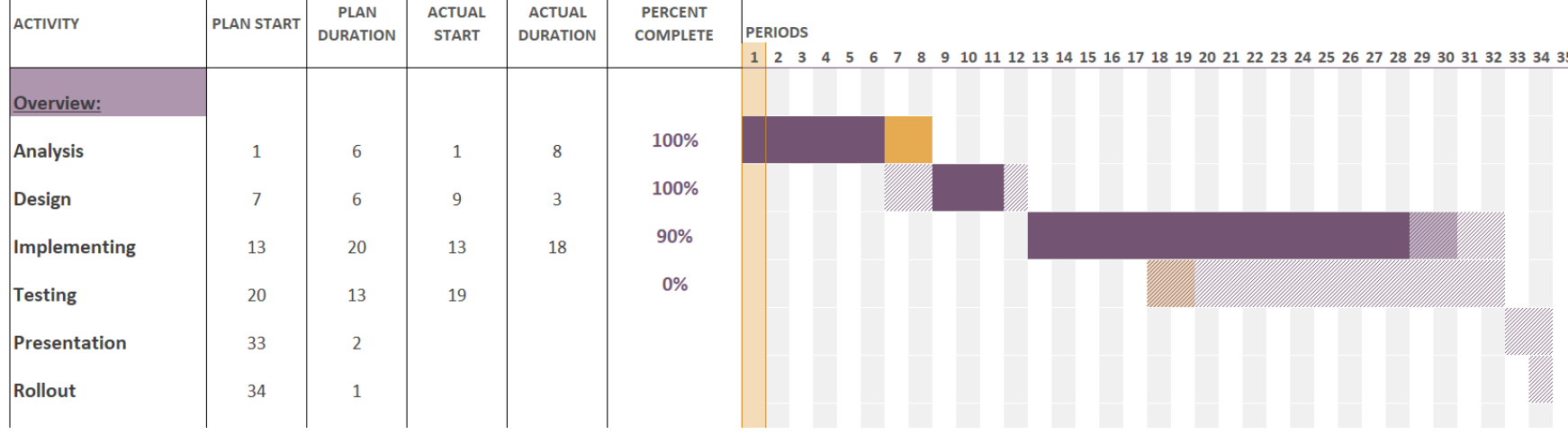
The **Testing phase** ensures system quality through structured validation. This includes the development of a Software Test Plan (STP), execution of black-box, white-box, and unit tests, as well as systematic error documentation.

Finally, the **End Phase** concludes the project with the formal handover to stakeholders, ensuring that all deliverables, documentation, and results are completed and ready for presentation.

5. Milestones

Milestone	Description	Planned Week
Requirement Clarification BC	Final clarification and sign-off of BC, CRS, and AAS integration	6
Design Approval	Completion and acceptance of wireframes, SRS, SAS	12
Code Implementation	All major functionalities implemented	32
Testing & Error Documentation	All tests completed and errors documented	32
Project Handover & Presentation	Delivery to stakeholders and presentation	34

6. Project Timeline



As can be seen in the Gantt chart of the **Business Case (BC)**, the project is planned over a total of **37 working weeks**, distributed as follows:

- Analysis:** 6 weeks
- Design:** 6 weeks
- Implementation:** 20 weeks
- Testing:** 13 weeks
- End Phase:** 3 week

7. Risks

This section identifies potential risks that may affect the successful execution of the project.

For each risk, possible impacts are considered and appropriate mitigation strategies are defined.

#	Risk	Mitigation / Countermeasure
1	Dependencies change (e.g., BaSyx upstream)	Early detection, include time buffers, regular rebasing
2	Migration / integration errors	Rollback strategy, backups, incremental integration
3	Emerging security vulnerabilities	Code reviews, static analysis, alternative libraries if needed
4	Team members unavailable	Redundant task distribution, knowledge documentation
5	Schedule overrun	Early communication with stakeholders, reprioritization
6	Backup failure	Decentralized backups (GitHub + local + cloud)
7	Communication problems	Weekly meetings, asynchronous updates, clarify uncertainties

The project team will actively monitor and manage these risks. Each risk is regularly reviewed, and mitigation strategies are applied throughout the project lifecycle. The **Risk Matrix** visually summarizes risk probability and impact.

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